

Enhancing VR Customer Service Training: A System for Generating Customer Queries and Evaluating Trainee Responses

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ABSTRACT

Virtual customer service training using avatars eliminates the need for physical facilities, reducing costs and enabling remote participation. However, the challenge lies in the authoring cost of preparing service specific question and the necessity of experienced instructors. To address this, we employ LLM technology to generate service specific question and evaluate trainee's answer mitigating authoring costs. Despite LLMs having general knowledge limitations, we integrate RAG to imbue them with specific service knowledge. This innovation facilitates the generation of service specific questions. Furthermore, we are developing a self-training system where RAG assesses the correctness of trainee answers, enabling independent learning (without an instructor). This paper discusses the issues faced and insights gained during the development of our system.

CCS CONCEPTS

• **Software and its engineering** → **Software prototyping.**

KEYWORDS

VR training, AI, avatar, large-scale language model, LLM, RAG, Retrieval Augmented Generation

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1 INTRODUCTION

Customer service requires exceptional communication skills, and various training methods like lectures and role-playing are used to enhance these skills. Traditional training, however, involves significant costs and resource demands, such as facilities and supervisors. Recently, VR training systems have shown promise for customer

service skill development [Carruth 2017] [Colbert et al. 2016]. We are currently developing a commercial VR system for this purpose.

VR training faces challenges like high authoring costs and the need for experienced supervisors to provide feedback. The expenses for creating virtual spaces and customer avatars are significant. Solutions such as Luma AI's "Genie" and "Interactive Scenes" help reduce these costs by generating 3D models and environments from text and videos [AI 2024a]. Additionally, we propose using LLMs-based algorithms to enable autonomous responses from customer avatars [Hara et al. 2024]. However, conventional LLMs struggle with generating service-specific questions, and supervisor feedback is still needed. For commercial viability, it's crucial to enable trainees to practice independently.

To address this, we are developing a system capable of automatically generating service-specific questions and assessing trainees. This paper outlines our findings and challenges encountered in constructing a prototype system that generates tailored questions and evaluates the accuracy of trainee responses.

2 TRAINEES EVALUATION PROTOTYPE SYSTEM

We employed LLMs combined with Retrieval-Augmented Generation (RAG) to develop a framework where customer avatars autonomously ask targeted questions about services and evaluate trainee responses. LLMs typically lack specific service knowledge, making precise question formulation difficult. RAG addresses this by sourcing facts from external knowledge repositories, aiding LLMs in generating specific questions and assessing response accuracy. Our RAG implementation follows a structured seven-step approach, as depicted in Figure 1. We implemented RAG using LangChain[LangChain 2024] in the following 7 steps (Figure 1 also shows 7 steps). In this time, we use "ChromaDB[ChromaDB 2024]" as the VectorDB and "GPT4ALL[GPT4ALL 2024]" and "gpt4all-falcon-q40.gguf[AI 2024b]" model as the LLM.

- (1) Collect service specific information (e.g. Hotel's hot springs, Yukata(Pajamas), Breakfast).
- (2) Retrieve and Embed service specific information to VectorDB.
- (3) Randomly select the topic of specific services (e.g. hot springs)
- (4) Perform a similarity search for hot springs on VectorDB and extract the top 10 results.

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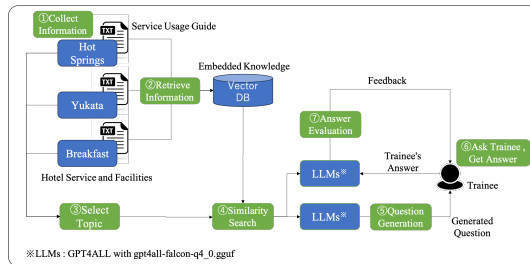
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Table 1: Hot Springs Information

Hot springs for both men and women are located on the basement floor of our hotel. Usage hours are 14:00-24:00 and 1:00-9:30. The hotel has 5 elevators. The elevator with the words "Onsen Gokuraku no Yu" written on the door goes to the hot spring on the first basement floor. Please note that other elevators do not go to the 1st basement floor. When you get off the elevator, you will find the hot spring reception counter. Please take off your shoes, put them in the shoe box, present your room key, and take the towel from the counter.

- (5) Pass the extracted 10 results to LLM and generate questions regarding the hot springs.
- (6) Ask the trainee the generated questions
- (7) Pass the extracted 10 results to LLM and evaluate whether the trainee's answers are accurate.

**Figure 1: Question generation and answer evaluation flow based on expert knowledge**

3 RESULTS, DISCUSSION, AND FUTURE WORKS

Table 1 presents the hotel-specific hot spring service information integrated into RAG. Additionally, Table 2 showcases examples of generated questions, trainee responses, and the subsequent evaluation thereof.

The results demonstrate that RAG effectively generates precise inquiries about hotel-specific hot springs and accurately evaluates trainee responses. We assessed the "Evaluation Correctness" using 50 datasets of generated questions, responses, and evaluations, rated by 10 individuals, averaging a score of 4.56. We used the same LLMs and models for both question generation and answer evaluation in this study. Moving forward, we aim to enhance accuracy by employing purpose-specific LLMs and models separately. One current challenge is the 4 to 5 seconds required for question generation, which we plan to optimize. Our research focuses on developing a system that autonomously generates tailored service-related questions using RAG, equipping LLMs with extensive service domain knowledge. We're also creating a system for trainees to independently assess their answers' correctness. Future efforts will include evaluating this system in various customer service settings, such as supermarkets and airports, to quantitatively measure its utility and effectiveness.

Table 2: The example of Generated Question, Trainee's answer and answer evaluation about Hot Springs.

Generated Question	Trainee's Answer	Answer Evaluation with RAG
The operating hours are 14:00-24:00 and 1:00-9:30, but is it possible to use the hot springs even late at night?	Yes, but it will not be available from 24:00 to 1:00 due to cleaning.	The employee's answer matches the hotel's description of the hot springs, and I appreciate that they provide accurate information. It is also properly communicated that the property is not available during late night hours (24:00 to 1:00) due to cleaning.
Regarding the use of the hot springs, what specific procedures should I follow after taking the elevator to the 1st basement floor?	You'll understand if you go to a hot spring	In this case, your answer may not be enough and difficult for the guest to understand. It is important to provide information to guests in an easy-to-understand manner.

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